

Challenge: Replace Outliers with Median

Outliers can significantly impact the quality of your data analysis, especially when they arise from errors or rare events that do not reflect typical patterns. When you want to reduce the influence of extreme values while keeping all your data points, replacing outliers with the **median** of the column is a robust technique. The **median** is resistant to the effect of outliers, so it provides a stable replacement value that maintains the overall distribution of your data. This approach is especially useful when you want to avoid losing data by removing rows, and when the **mean** would be skewed by the very outliers you are trying to address.

```
import pandas as pd

# Example DataFrame with outliers in the 'score' column
data = {
    "name": ["Alice", "Bob", "Charlie", "David", "Eve"],
    "score": [85, 90, 300, 88, 92] # 300 is an outlier
}
df = pd.DataFrame(data)

# Let's say outliers have been identified using the IQR method
# For this example, we know that 300 is an outlier
outlier_mask = df["score"] > 150

print("Original DataFrame:")
print(df)
print("\nOutlier mask:")
print(outlier_mask)
```